

CHAOYANG, China

WMO: 543240

Lat: 41.5478N

Lon: 120.4292E

Elev: 176

StdP: 99.23

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.9	-16.7	-29.9	0.2	-11.5	-27.9	0.3	-11.0	7.5	-5.1	7.0	-5.1	1.4	180	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	34.2	21.3	32.5	21.3	31.2	21.0	25.6	30.1	24.7	28.8	23.9	27.7	3.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.3	19.7	28.3	23.5	18.7	27.4	22.7	17.8	26.5	80.4	30.2	76.5	28.9	72.9	27.6	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
7.9	6.9	6.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-21.1	37.0	2.9	2.4	-23.1	38.7	-24.8	40.1	-26.4	41.4	-28.5	43.2
			-21.6	26.7	2.7	0.9	-23.6	27.3	-25.1	27.8	-26.6	28.3	-28.6	28.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-8.2	-3.9	3.5	12.2	18.9	23.1	25.6	24.2	18.9	11.1	1.5	-5.9
	DBStd	12.52	4.17	4.75	5.10	4.90	4.06	3.11	2.32	2.67	3.40	4.57	5.35	4.17
	HDD10.0	1988	563	390	211	32	1	0	0	0	0	42	258	492
	HDD18.3	3654	821	623	459	191	41	2	0	1	33	227	505	751
	CDD10.0	2048	0	0	9	99	277	393	482	441	268	76	3	0
	CDD18.3	673	0	0	0	8	59	146	224	183	51	3	0	0
	CDH23.3	6354	0	0	2	98	682	1364	2098	1612	470	28	0	0
	CDH26.7	2349	0	0	0	20	244	558	824	589	110	4	0	0
Wind	WSAvg	2.5	2.1	2.5	3.2	3.5	3.2	2.7	2.5	2.1	2.1	2.3	2.3	2.1
Precipitation	PrecAvg	478	2	3	5	21	36	76	163	102	45	17	6	2
	PrecMax	687	10	19	31	85	84	139	377	294	135	75	20	20
	PrecMin	260	0	0	0	1	5	12	53	18	0	0	0	0
	PrecStd	114	3	4	8	17	20	33	92	60	28	17	6	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	12.0	22.4	29.3	34.8	36.8	36.4	35.4	31.2	26.3	18.3	8.1
		MCWB	0.1	3.9	9.6	14.9	17.2	19.1	23.4	22.5	19.4	15.6	8.9	2.6
	2%	DB	3.1	9.4	18.1	25.9	31.6	33.7	33.9	32.9	29.4	23.5	14.9	5.2
		MCWB	-2.1	2.2	7.8	12.5	16.3	19.6	23.2	22.9	18.7	13.9	7.5	0.3
	5%	DB	1.0	6.9	14.9	23.6	29.4	31.9	32.3	31.3	27.8	21.3	12.5	3.2
		MCWB	-3.3	0.8	5.9	11.4	15.5	19.0	23.0	22.5	17.7	12.6	6.3	-1.3
	10%	DB	-0.9	4.6	12.3	21.1	27.2	30.0	30.9	29.9	26.2	19.1	10.0	1.2
		MCWB	-4.7	-0.7	4.5	10.2	14.6	18.7	22.7	22.2	17.2	11.5	4.5	-2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.4	4.6	10.4	16.3	20.3	23.8	26.8	26.5	22.7	17.7	10.7	3.6
		MCDB	5.5	10.3	21.0	25.6	29.9	29.2	31.9	31.5	28.7	23.2	15.2	7.1
	2%	WB	-1.7	2.7	8.4	14.2	18.5	22.2	26.0	25.5	21.1	15.7	8.2	0.5
		MCDB	2.8	8.4	16.6	22.9	27.7	27.7	30.8	29.7	25.8	20.9	13.9	4.7
	5%	WB	-3.2	1.0	6.5	12.6	17.2	21.4	25.1	24.6	20.0	14.3	6.6	-1.2
		MCDB	0.9	6.6	14.5	21.2	25.6	26.9	29.5	28.5	25.0	19.5	11.9	2.9
	10%	WB	-4.7	-0.6	4.7	11.1	16.1	20.5	24.3	23.7	18.9	12.7	4.8	-2.7
		MCDB	-1.0	4.4	11.6	19.3	23.9	26.1	28.3	27.5	23.5	17.8	9.6	1.1
Mean Daily Temperature Range	5% DB	MDBR	11.7	11.9	12.0	12.2	12.4	10.8	9.2	10.0	12.5	12.3	10.8	10.9
		MCDBR	13.9	14.4	15.7	15.7	16.0	14.1	11.9	12.0	14.5	14.8	13.9	12.8
		MCWBR	10.0	9.5	8.8	7.2	6.0	4.6	4.0	4.3	5.5	7.1	8.2	9.1
	5% WB	MCDBR	13.5	13.7	14.7	13.4	13.0	10.4	8.8	9.0	11.2	12.5	13.0	12.5
		MCWBR	9.9	9.3	8.3	6.8	5.7	4.2	4.0	4.2	5.3	6.9	8.2	9.0
Clear-Sky Solar Irradiance	taub	0.338	0.405	0.457	0.556	0.609	0.705	0.684	0.595	0.533	0.499	0.400	0.348	
	taud	2.163	1.984	1.898	1.728	1.669	1.554	1.640	1.799	1.863	1.861	2.043	2.150	
	Ebn at Noon	780	769	776	727	701	635	644	689	698	658	689	727	
	Edh at Noon	101	143	175	223	242	273	248	206	180	160	114	95	
All-Sky Solar Radiation	RadAvg	2.50	3.39	4.58	5.35	5.99	5.39	5.13	5.04	4.54	3.45	2.48	2.15	
	RadStd	0.16	0.22	0.29	0.33	0.47	0.46	0.46	0.35	0.25	0.20	0.22	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (6 neighbors)	N/A	N/A	-0.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page